

ChE 382: Unit Operations Laboratory

PROCESS CONTROL: LEVEL CONTROL

Revision: Spring 2019v1

Introduction

The Armfield PCT50 Level Control process unit provides a compact, hands-on introduction to the fundamental principles of process control engineering. The fluid-based process provides a clear demonstration of a single PID loop using Level as the measured variable. The effect of making changes to the system or to the controller configuration can be quickly investigated by applying repeatable disturbances to the process, clearly demonstrating the need for correct matching of system characteristics and controller settings. The software allows all appropriate measurements and controller settings to be displayed continuously and recorded for later analysis, if required by the user. Alternatively the process can be operated using an external controller such as the optional Industrial PID controller (PCT54) or Programmable Logic Controller (PCT55) (**we will not use the PID or PLC**).

PCT50 Level Control Process

Refer to the equipment diagram in the Armfield software application for details about the PCT50 Level Process. The front view shows the layout of the PCT50 and the circuit diagrams show the process in operation with inflow to the process vessel (controlled by varying pump speed) or outflow from the process vessel (controlled by opening and closing a solenoid valve on a time cycle). We will be using the operation with inflow for our experimental procedure.

PCT50 is a level control process using water as the working fluid for safety and convenience in use. Water stored in the lower sump tank is transferred to the upper process vessel via a submerged, variable speed centrifugal pump (10). A quick release connector (9) allows the flexible pump outlet tube to be disconnected to aid priming of the pump after filling the sump tank with water. A vertical inlet arrangement in the process vessel allows visualization of the water entering the vessel, independent of water level and an integral non-return valve (8) prevents water from draining back into the sump vessel when the pump speed is reduced or stopped. An inline ball valve (CV1) above the quick release connector allows the flow of water entering the process vessel to be varied to suit a particular demonstration.

The water level inside the process vessel is measured using an electronic pressure sensor (2) mounted through the floor of the vessel. One side of the pressure sensor is connected to the inside of the process vessel and the other is open to atmosphere, allowing the pressure in the process vessel to be measured relative to atmosphere. This sensor therefore measures the level of water inside the process vessel. The level is indicated (1) on the scale on the side of process vessel.

Water drains from the process vessel back into the lower sump tank via two outlets in the floor of the process vessel. Flow through the main outlet is continuous. Flow through the second outlet can be started and stopped by a solenoid valve (SOL) under remote control. Both outlets incorporate a hand operated ball valves (CV2 and CV3) that allows the flow of water to be varied continuously to suit a particular demonstration. Both outlets also incorporate an interchangeable orifice (3 and 5) that allows the flow to be fixed at predetermined values, varying only with the size of the orifice and the head in the process vessel. The orifice size is changed by unscrewing the plastic fitting containing the orifice and replacing it with the required alternative. The fitting incorporates an 'O' ring seal and only requires hand

tightening. The alternative sizes of orifice (12) are stored in threaded holes at the front of the base plate when not in use.

An overflow (6) in the process vessel returns water to the sump tank, preventing the process vessel from flooding in use.

The pressure sensor measuring the level, the centrifugal pump and the solenoid valve are connected to an electrical interface (4) that incorporates the necessary signal conditioning, allowing the process to be operated directly from a PC using a single USB port. Alternatively the process can be operated using a controller such as PCT54 (PID Controller) or PCT55 (Programmable Logic Controller) connected to the electrical interface.

The computer software supplied with PCT50 allows control of the level process and data logging of the responses using a PC. Alternatively the software allows data logging only while operating the process remotely using a PID controller or PLC.

The unit is drained using the ball valve (11) located at the rear.

Initial Filling the Sump Tank

No permanent supply of water is required when operating the PCT50. Ensure that the drain valve at the rear of the sump tank is fully closed before filling the sump tank. The sump tank can be filled using a hose or by pouring water through the cut out at the front or via the process vessel at the top. The water level should be approximately 30 mm below the cut out in normal use. For safe handling the sump tank should be drained before moving the equipment to a different location.

Note: To ensure reliable priming of the water pump after filling for the first time or after draining and refilling the equipment, the flexible tube from the pump outlet can be temporarily disconnected from the system by pressing the latch on the quick release connector. After running the pump briefly at high speed, with the tube disconnected to expel any trapped air, the tube can be reconnected for normal operation.

EXPERIMENTS:

Experimental Procedure:

Exercise A and Exercise C will be performed using the PCT50 Level Process Unit. See following pages for the experimental procedures. Refer to the Manual to adjust the data sampling rate to obtain a reasonable amount of data per Exercise (default value creates a huge amount of data points).

EXERCISE C: Follow the procedure to the bottom of page 5. But if time permits, conduct further experiments at different values of the P-term to better understand the relationship between the P and I terms.

Results and Discussion:

Exercise A: Make a plot of your results from your manual changes. Also tabulate your data: Level, Pump Speed, Operation (Initial fill, solenoid on/off), Time. Briefly describe why the behavior is occurring due to the disturbance. How can you use the information to incorporate proportional control?

Exercise C:

P Only Controller: Make plots of your results for the Proportional gain experiments for different values of the P term. Discuss any offset from the setpoint value and stability of P-only control. Analyze your results and discuss the setting you recommend as giving the most stable control.

P+I Controller: Make plots of your results for the proportional+integral control at different I settings. Explain and discuss the behavior at the different I settings, and if performed, at different P settings. How does the P and I term influence each other?

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Exercise A - Manual control of water level (Open loop)

Objective

To control the level of water in the process vessel by manually varying the speed of a pump supplying water to the process vessel (inflow control).

To determine the characteristics of the level process with different ranges of inflow and outflow to / from the process vessel.

Method

Using a PC to operate the process, the pump speed (water flow into the process vessel) will be varied by the operator in an attempt to maintain a steady water level in the process vessel.

With outflow valve CV2 fully open water will drain from the process vessel continuously, the rate determined by the size of the orifice fitted and the depth of water in the vessel.

Disturbances will be applied to the process by opening and closing the solenoid valve. This changes the outflow from the process vessel resulting in a change in the level inside the vessel. The magnitude of the disturbance is determined by the size of the orifice fitted below the solenoid valve and the depth of water in the vessel.

A pressure sensor measures the water level in the process vessel and indicates the value in mm depth of water on the mimic diagram.

Equipment Required

PCT50 Level Process

PC with PCT50 software loaded

Theory

For the level to remain constant in the process vessel the flowrate into the vessel supplied by the pump must match the flowrate leaving the vessel via CV2 and the solenoid valve, if open. Since the flow of water entering the vessel is proportional to pump speed, the speed must be increased to raise the level and decreased to lower the level.

Equipment set up

Ensure that the apparatus has been set up according to the Installation section, the sump filled with clean water and the power supply connected to the socket marked 24V IN at the rear of the electrical enclosure.

Before starting this exercise fully open outlet valve CV2 and fit a 4 mm diameter orifice below it. Fit a 2 mm orifice below the solenoid valve.

Connect the USB socket at the rear of the electrical enclosure a suitable PC on which the PCT50 software has been installed using the USB cable supplied. Check that the PC is switched on then run the PCT50 software and select Experiment Ex1: Inflow Control (PC Control).

Ensure that the Virtual COM port has been enabled using the red telephone icon in the top toolbar.

Before starting the exercise it is advisable to adjust inlet valve CV1 to give appropriate flow into the process vessel as follows:

Fully open outlet valve CV2 and fully open inlet valve CV1.

Choose the  icon (or View \ Diagram) to display a mimic diagram of the process.

Click Power On (in Controls).

If the level indicated on the mimic diagram is not 0mm with the vessel empty click the zero button to zero the reading.

Set the pump speed to 50%, by typing in the value or by clicking the up arrow, so that water is pumped into the process vessel.

As the water level rises adjust inlet valve CV1 until the level remains steady at approximately mid height in the process vessel (flow into the vessel via CV1 matches flow out of the vessel via CV2).

Reset the pump speed to 0 % on the mimic diagram and allow the vessel to drain via outlet valve CV2.

Procedure

Gradually increase the speed of the pump and observe that the level rises in the process vessel when the flow into the vessel exceeds the flow out. Observe that the level only remains constant if the pump speed is adjusted to match the outflow. If the pump speed is too high then the level will continue to rise, if the pump speed is too low then the level will continue to fall.

With the level constant in the process vessel choose the  icon to begin data logging and note the level reading L1 in mm.

Open the solenoid valve by clicking the button on the mimic diagram (1 indicates that the valve is open). Observe that the level falls slowly due to the additional outflow via the solenoid valve. When the level has fallen by at least 20 mm note the current pump speed indicated on the mimic diagram then increase the pump speed to restore the original level.

Close the solenoid valve by clicking the button on the mimic diagram (0 indicates that the valve is closed).

Observe that the level rises very slowly because the flow from the pump now exceeds the flow from the base of the process vessel. When the level has risen by at least 20 mm note the current pump speed indicated on the mimic diagram then reduce the pump speed to restore the original level.

Choose the  icon to finish data logging.

Choose the  icon (or View \ Graph) to display a graph of the response obtained. Observe the rising and falling water level as the solenoid is opened and closed and action is taken to restore the level.

Return to the mimic diagram, reduce the pump speed and allow the process vessel to drain.

Create a new results sheet by selecting the  icon in the tool bar of the software then choose the  icon to begin data logging.

Gradually increase the pump speed until the level just starts to rise. Leave the pump at this speed and allow the vessel to fill. When the level reaches 150 mm stop the pump and allow the vessel to drain.

When the vessel is empty start the pump at 75 % speed and allow the vessel to fill. When the level reaches 150 mm stop the pump and allow the vessel to drain.

Choose the  icon to finish data logging.

View the graph and observe the different rates of filling at different pump speeds. Also note that the vessel drains at the same rate, defined by the outlet orifice below valve CV2. These characteristics will affect the control of the process.

If time permits repeat the exercise with different sizes of orifice fitted to the outlet valve CV2 and solenoid valve, E.g. 2 mm at CV2 and 4 mm at the solenoid valve (smaller outflow with larger disturbance). The filling characteristics with different settings of CV1 could also be investigated.

Create a new results sheet for each run by selecting the  icon in the tool bar of the software.

Results

Save a copy of the results obtained so that the data can be viewed in graphical or tabular format at a later date.

Conclusion

The level in a vessel will only remain constant when the flow into the vessel matches the flow leaving the vessel.

Excess water entering the vessel will cause the level to rise. Insufficient water entering the vessel will cause the level to fall.

When level is stable in a vessel, with inflow matching outflow, any disturbance such as additional flow entering or leaving the vessel will result in a change in level, requiring variation of the inflow to restore the level.

It is important that the flow from the pump is suitable over the available range of pump speeds to match the outflow from the process.

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Exercise C - Proportional control of water level (Closed loop, P only and P + I)

Objective

To control the level of water in the process vessel using a proportional controller to automatically vary the speed of the feed pump (inflow control).

To determine the response of a level process when using a P only controller to vary the pump speed.

To determine the response of a level process when using a P+I controller to vary the pump speed.

To change the level inside the process vessel by changing the set point on the P or P+I controller and to determine the effect of disturbances to the process.

To investigate the effect of different ranges of inflow and outflow to / from the process vessel.

Method

- Using a PC to operate the process, the pump speed (water flow into the process vessel) will be varied automatically by a proportional controller in an attempt to maintain a steady water level in the process vessel.

The effect of the Proportional Band setting (P) can be demonstrated by configuring a P only controller and observing the response of the process to different settings of P.

The contribution of the ~~control~~ Integral Time setting (I) can be demonstrated by configuring a P+I controller and observing the response of the process to different settings of I with P fixed then different combinations of P+I.

With outflow valve CV2 fully open water will drain from the process vessel continuously, the rate determined by the size of the orifice fitted.

Disturbances can be applied to the process by opening and closing the solenoid valve. This changes the outflow from the process vessel resulting in a change in the level inside the vessel. The magnitude of the disturbance is determined by the size of the orifice fitted below the solenoid valve.

A pressure sensor measures the water level in the process vessel and indicates the value on the PC.

Equipment Required

PCT50 Level Process.

PC with PCT50 software loaded

Theory

Proportional term (P)

The Proportional Band setting on a process controller makes a change to the output (pump speed on PCT50) that is proportional to the current error value (the difference between the measured level and

the set point on the controller). The proportional response can be adjusted by multiplying the error by a constant K_p , called the Proportional Gain. This is related to the Proportional Band setting on the controller as follows:

$$\text{Proportional Gain } (K_p) = 100 \% / \text{Proportional Band } \%$$

i.e. 100 % PB means unity gain (change in controller output = error at input)

and 50 % PB means a gain of 2 (change in controller output = 2x error at input)

A low setting of the Proportional Band (large gain) results in a large change in the output for a given change in the error. If the Proportional Band is too low, the system can become unstable. In contrast, a large setting of the Proportional Band (low gain) results in a small output response to a large input error, and a less responsive or less sensitive controller. If the Proportional Band is too high, the control action may be too small when responding to system disturbances resulting in slow response and offsets of the resulting process variable from the set point.

A Proportional-only controller will not always settle at the set point, but may retain a steady-state offset. Offset can be reduced in Proportional-only control by reducing the Proportional Band setting. However, if the PB is set too small then hunting or oscillating will occur. The offset can be minimised by adding a bias to the set point (setting the set point above or below the required value to compensate for the offset) but this technique is only appropriate if the system characteristics are known and fixed. A better solution is to remove the offset by adding Integral action to the controller (P+I) as described below.

Integral term (I)

The contribution from the integral term is proportional to magnitude and duration of the error. The Integral term in a PID controller is the sum of the instantaneous error over time and gives the accumulated offset that should have been corrected previously. The resulting controller output is the sum of the contribution from the Integral term and the contribution from the P term.

When the I term is correctly adjusted any residual offset in the process variable due to the P term will be gradually reduced by the Integral term until the offset is eliminated. If the time setting of the I term is too long then correction to any offset will be very slow.

However, since the integral term responds to accumulated errors from the past, it can cause the present value to overshoot the setpoint value or to make the process completely unstable if the time setting of the I term is too short. If this occurs the I term makes adjustments to the controller output faster than the process can respond. I.e. the I term winds up the controller output so that the process overshoots considerably, hence the term Integral Wind-up or Integral saturation.

Careful selection of the I term in combination with the P term will give efficient response to changes in the system.

Derivative term (D)

The derivative term slows the rate of change of the controller output. Derivative control is used to reduce the magnitude of the overshoot produced by the I term and improve the combined controller-process stability. However, the derivative term slows the response of the controller. Also, differentiation of a signal amplifies noise and thus this term in the controller is highly sensitive to noise in the error term, and can cause a process to become unstable if the noise and the derivative gain are sufficiently large.

Equipment set up

Ensure that the apparatus has been set up according to the Installation section, the sump filled with clean water and the power supply connected to the socket marked 24V IN at the rear of the electrical enclosure.

Before starting this exercise fully open outlet valve CV2 and fit a 4 mm diameter orifice below it. Fit a 2 mm orifice below the solenoid valve.

Connect the USB socket at the rear of the electrical enclosure a suitable PC on which the PCT50 software has been installed using the USB cable supplied. Check that the PC is switched on then run the PCT50 software and select Experiment Ex1: Inflow Control (PC Control).

Ensure that the Virtual COM port has been enabled using the red telephone icon in the top toolbar.

Before starting the exercise it is advisable to adjust inlet valve CV1 to give appropriate flow into the process vessel as follows:

Fully open outlet valve CV2 and fully open inlet valve CV1.

Choose the  icon (or View \ Diagram) to display a mimic diagram of the process.

Click Power On (in controls).

If the level indicated on the mimic diagram is not 0mm with the vessel empty click the zero button to zero the reading.

Set the pump speed to 50%, by typing in the value or by clicking the up arrow, so that water is pumped into the process vessel.

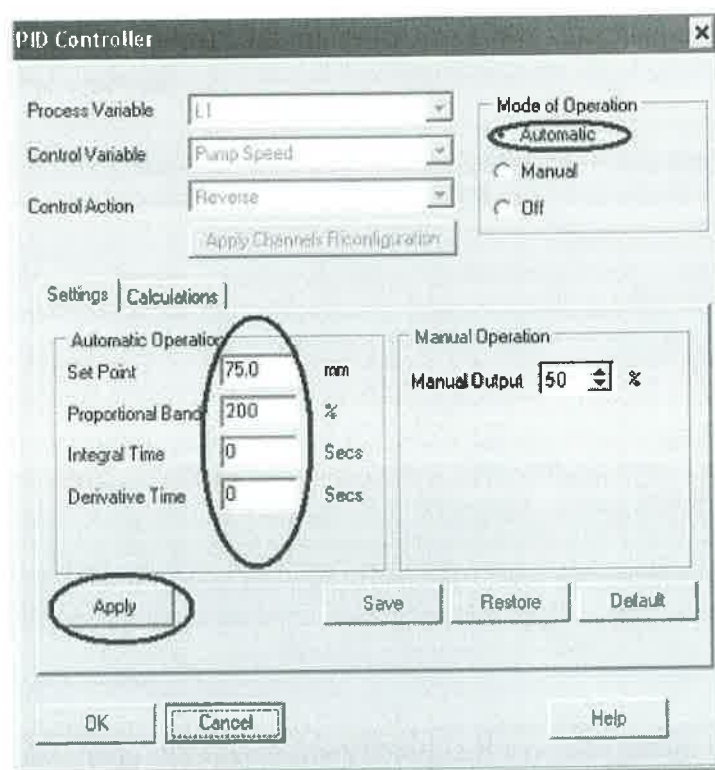
As the water level rises adjust inlet valve CV1 until the level remains steady at approximately mid height in the process vessel (flow into the vessel via CV1 matches flow out of the vessel via CV2).

Reset the pump speed to 0% on the mimic diagram and allow the vessel to drain via outlet valve CV2.

Procedure - P only controller

Choose the PID box on the mimic diagram, set the Proportional Band P to 200%, the Integral Time I to 0 and the Derivation time D to 0 then set the Set Point to 75 mm (default at start up). Click Apply to enter the changes to the settings.

Choose the Automatic Mode of Operation.



Select the  icon to begin data logging.

The pump will start, the level will gradually rise and water will drain from the vessel via outlet valve CV2. The level will continue to rise until it reaches a steady level but this will not correspond with the setpoint of 75 mm on the controller because of the offset inherent in a P only controller.

Open the solenoid valve by clicking the button on the mimic diagram (1 indicates that the valve is open) to disturb the process. Observe that the level falls slightly, the pump speeds up slightly due to the increased error but a large offset remains. When the level has settled close the solenoid valve and allow the level to settle.

Adjust the set point to 100 mm then click apply and observe the response to a requested increase in water level. When the level has settled, return the set point to 75 mm and observe the response.

Adjust the P term to 100 % in the PID controller then click Apply. Observe that the pump speeds up and the offset reduces slightly. Open then close the solenoid valve as before to disturb the process and observe the response, the stability of the level and any reduction in offset.

Adjust the P term to 50 % and repeat the above steps.

Adjust the P term to 20 % and repeat the above steps.

Adjust the P term to 10 % and repeat the above steps

Adjust the P term to 5 % and repeat the above steps

Adjust the P term to 2 % and repeat the above steps.

Choose the  icon to finish data logging.

Results – P only controller

Choose the  icon (or View \ Graph) to display a graph of the responses obtained.

Choose Format \ Graph Data and Plot Level L1 and Set point on 1 axis, and Controller Output on the second axis.

Observe the changes in water level and the stability of the responses for each change in P setting with a disturbance or change in set point.

Choose a setting for P that gives stable control with a reasonably small offset that can be used in the next part of the exercise introducing Integral action.

Procedure - P+I Controller

Create a new results sheet for this run by selecting the  icon in the tool bar of the software.

Choose the PID box on the mimic diagram, set the Proportional Band P to the value selected above, the Integral Time I to 0 seconds and the Derivation time D to 0 then set the Set Point to 75 mm. Click Apply.

Select the  icon to begin data logging.

Choose the Automatic mode of operation and allow the water level to stabilise.

Adjust the I term to 100 seconds in the PID controller then click Apply. Observe that the pump speed varies to reduce the offset. When the level is settled at the set point open the solenoid valve to disturb the process. Observe that the level falls slightly, the pump speed varies and the level gradually returns to the set point. When the level has settled close the solenoid valve and allow the level to settle.

Adjust the set point to 100 mm then click Apply and observe the response to a requested increase in water level. When the level has settled return the set point to 75 mm and observe the response.

Adjust the I term to 50 seconds in the PID controller then click Apply. Observe that the level returns to the set point faster than before. When the level is settled at the set point open then close the solenoid valve as before to disturb the process.

Adjust the set point to 100 mm then click Apply and observe the response to a requested increase in water level. When the level has settled return the set point to 75 mm and observe the response.

Continue reducing the Integral time setting and repeating the disturbance and set point changes until the level becomes unstable.

Results – P+I controller

Choose the  icon (or View \ Graph) to display a graph of the responses obtained.

Choose Format \ Graph Data and Plot Level L1 and Set point on 1 axis, and Controller output on the second axis.

Observe the changes in water level and the stability of the responses for each change in I setting with a disturbance or change in set point.

Choose a setting for I that gives stable control and reduces any offset swiftly without causing the system to become unstable.

Since the effect of Integral Action is related to the Proportional Band setting (reduced PB gives greater effect for any setting of I) the above exercise should be repeated at a completely different value of PB to determine the necessary change in I for effective control of the level and to demonstrate the relationship between P and I.

If time permits repeat the exercise with different sizes of orifice fitted to the outlet valve CV2 and solenoid valve, E.g. 2 mm at CV2 and 4 mm at the solenoid valve (smaller outflow with larger disturbance). The filling characteristics with different settings of CV1 could also be investigated e.g. partly closing valve CV1 to reduce the flow or increasing the flow by opening the valve.

Conclusion

A large proportional band setting will result in very stable control but will result in slow changes and large offsets in the process variable.

Reducing the proportional band will improve the speed of response and give smaller offsets but the process may become unstable if the proportional band is reduced too far.

- * Optimum results are obtained with the lowest proportional band setting that gives stable control but with the addition of Integral action to eliminate any offset.

Reducing the Integral time setting increases the speed at which any offset is reduced but settings that are too short will result in instability because the process is unable to respond quickly enough to the changes from the controller. The effect is called integral saturation or integral wind up and must be avoided.

Increasing the PB reduces the effect of Integral Action so shorter setting of I is required to eliminate offset swiftly.